📝 Detailed Lecture Notes: Maximum Likelihood for the Binomial Distribution

Video: Maximum Likelihood for the Binomial Distribution, Clearly Explained!!!
Channel: StatQuest with Josh Starmer

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1. The Binomial Distribution and Its Parameters

The **Binomial Distribution** is used to model the number of **"successes" (X)** in a **fixed number of trials (n)**, where each trial has the same probability of success (P).

Binomial Probability Function [00:54]

The standard **Probability** function for the binomial distribution is:

$$P(X \mid n, P) = \binom{n}{X} \cdot P^X \cdot (1-P)^{(n-X)}$$

- X: The number of observed **successes** (e.g., people preferring orange Fanta).
- n: The total number of **trials** (e.g., total people asked).
- P: The underlying probability of success for **one trial** (the parameter we want to estimate).

Switching to Likelihood [02:15]

To use Maximum Likelihood Estimation (MLE), we switch the function's notation to **Likelihood (L)**. This signifies that the data (X and n) are **fixed** and we are solving for the unknown parameter (P):

$$L(P \mid n, X) = \binom{n}{X} \cdot P^X \cdot (1-P)^{(n-X)}$$

- The goal of MLE is to find the value of P that maximizes this likelihood function [04:32].

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2. Finding the Maximum Likelihood Estimate (MLE) for P

The process of finding the optimal P involves finding the point where the derivative (slope) of the likelihood function is equal to zero [04:51].

Step 1: Use the Log-Likelihood Function [05:11]

To simplify the mathematical process of differentiation (since the equation involves multiplication and exponents), we use the **logarithm of the likelihood function** (Log-Likelihood).

- This works because the maximum of the original function and the maximum of its logarithm occur at the same value of P [05:34].
- The logarithm converts **multiplication into addition** and **exponents into multiplication**, making the derivative much simpler [05:46].

Step 2: Take the Derivative and Set to Zero [06:18]

1. Take the derivative of the Log-Likelihood function with respect to the parameter, P.
2. Set the derivative equal to zero.
3. Solve the resulting equation for P.

Step 3: The Resulting MLE Formula [10:13]

The process of solving the derivative yields the **Maximum Likelihood Estimate (MLE)** for the probability of success, P:

$$\text{MLE for } \mathbf{P} = \frac{\mathbf{X}}{\mathbf{n}}$$

- **Interpretation:** The most likely value for the true underlying probability of success (P) is simply the **observed number of successes (X) divided by the total number of trials (n)** [10:23].
- While this result seems intuitive (it's the average success rate), the MLE process provides a **mathematical proof** that this intuitive estimate is indeed the value that maximizes the likelihood of observing the collected data [10:44].

http://googleusercontent.com/youtube_content/6